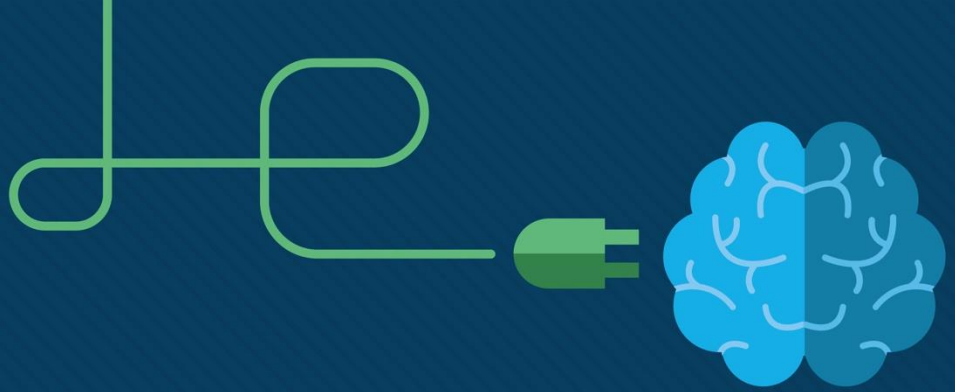

Module 7
Routing Concepts

ITNET03

Advanced Routing and Switching

Module Objectives

Module Title: Routing Concepts

Module Objectives:

- Describe the structure of a routing table.
- Explain static and dynamic routing concepts.
- Explain how routers select the best route and forward packets to the destination

Module References:

- CCNAv7 SRWE – Module 11

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7.1 IP Routing Table

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The Role of Routers

- **Recall:** A router is a network device that is used to move packets between logical networks

- The primary functions of a router:
  - Determine the best path to forward packets based on the information in its routing table
  - Forward packets toward their destination
- Routers also serve other beneficial functions as follows:
  - Broadcast containment by limiting their propagation within the local network.
  - Interconnect geographically separated locations.
  - Group users logically by application or department within an organization
  - Provide enhanced security by filtering unwanted traffic through access control lists.

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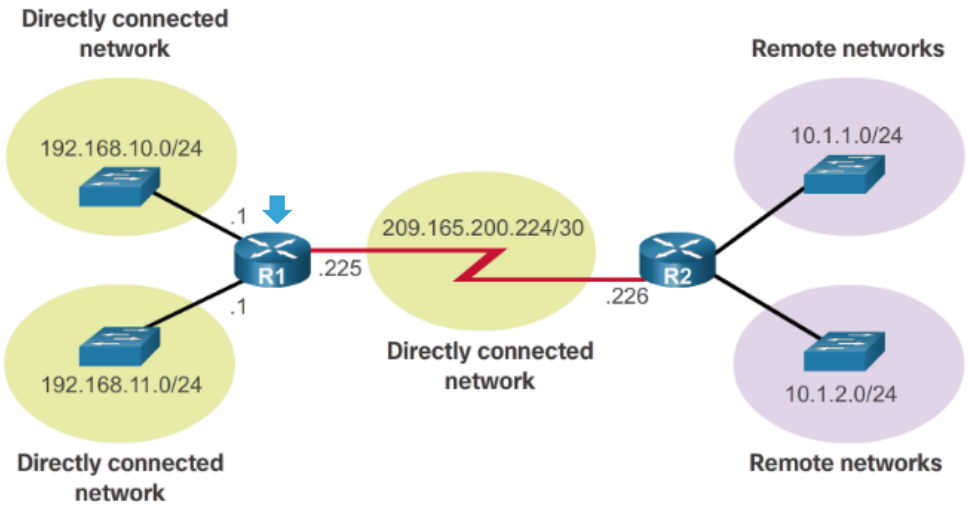
The Routing Table

- Each router maintains a routing table – a data structure stored in RAM to track the list of known networks and how to reach them.
- The routing table contains:
  - **Directly Connected Networks:** Networks that a router is a member of through one of its interfaces
  - **Remote Networks:** Networks that it must use a neighboring router to reach



The Routing Table

- The **show ip route and show ipv6 route** commands are used to display the contents of the IPv4 and IPv6 routing tables, respectively:

```
RTR-Main>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
...

Gateway of last resort is 192.168.1.1 to network 0.0.0.0

     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.12/32 is directly connected, GigabitEthernet0/0
     192.168.2.0/25 is subnetted, 2 subnets
S       192.168.2.0/25 [1/0] via 192.168.1.2, GigabitEthernet0/1.10
S       192.168.2.128/25 [1/0] via 192.168.1.2, GigabitEthernet0/1.10
     192.168.3.0/26 is variably subnetted, 2 subnets, 2 masks
D       192.168.3.0/26 [90/30720] via 192.168.1.13, 00:00:01, GigabitEthernet0/0
D       192.168.3.64/27 [90/30720] via 192.168.1.13, 00:00:01, GigabitEthernet0/0
S*   0.0.0.0/0 [1/0] via 192.168.1.1, GigabitEthernet0/0
```

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The Routing Table

- The **`show ip route`** **and** **`show ipv6 route`** commands are used to display the contents of the IPv4 and IPv6 routing tables, respectively:

```
R1# show ipv6 route
Codes: L - local, C - connected, S - static, R -
...
OE2 ::/0 [110/1], tag 2
   via FE80::2:C, Serial0/0/1
C 2001:DB8:ACAD:1::/64 [0/0]
   via GigabitEthernet0/0/0, directly connected
L 2001:DB8:ACAD:1::1/128 [0/0]
   via GigabitEthernet0/0/0, receive
C 2001:DB8:ACAD:2::/64 [0/0]
  via GigabitEthernet0/0/1, directly connected
L 2001:DB8:ACAD:2::1/128 [0/0]
   via GigabitEthernet0/0/1, receive
C 2001:DB8:ACAD:3::/64 [0/0]
   via Serial0/1/1, directly connected
L 2001:DB8:ACAD:3::1/128 [0/0]
   via Serial0/1/1, receive
O 2001:DB8:ACAD:4::/64 [110/50]
   via FE80::2:C, Serial0/1/1
O 2001:DB8:ACAD:5::/64 [110/50]
   via FE80::2:C, Serial0/1/1
L FF00::/8 [0/0]
   via Null0, receive
R1#
```

Routing Table Entries

IPv4 `D 201.165.224.0/24 [90/7680] via 192.168.1.1, 00:00:01, GigabitEthernet0/0`

IPv6 `O 2001:DB8:ACAD:4::/64 [110/50] via FE80::2:C, Serial0/1/1`

Routing table entries contain the following essential information about each known network

- **Route source** - This identifies how the route was learned.
- **Destination network (prefix and prefix length)** - This identifies the address of the remote network.
- **Administrative distance** - This identifies the trustworthiness of the route source.
- **Metric** - This identifies the value assigned to reach the remote network.
- **Next-hop** - This identifies the IP address of the next router to which the packet would be forwarded.
- **Route timestamp** - This identifies how much time has passed since the route was learned.
- **Exit interface** - This identifies the egress interface to use for outgoing packets to reach their final destination.

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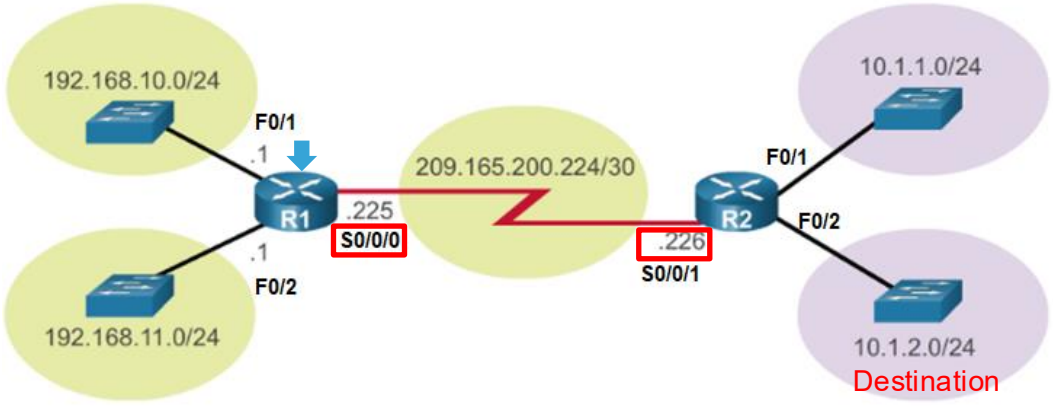
Next Hop and Exit Interfaces

IPv4 `D 201.165.224.0/24 [90/7680] via 192.168.1.1, 00:00:01, GigabitEthernet0/0`

IPv6 `O 2001:DB8:ACAD:4::/64 [110/50] via FE80::2:C, Serial0/1/1`

- The next hop and exit interfaces provide the router with the information needed to correctly forward a packet in the direction towards its final destination network
  - **Next-hop** identifies the **IP address of the next router** to which the packet would be forwarded to reach remote networks.
  - **Exit interface** identifies **this router's egress interface** to use to reach the destination network or the next router that can forward the packet

For **R1** to reach network 10.1.2.0/24:
Next Hop: **209.165.200.226**
Exit Interface: **S0/0/0**



Route Sources

A routing table obtains its routes to known networks from the following:

- From its own interfaces for directly connected networks
- From static routes and dynamic routing protocols for remote networks

The source for each route in the routing table is identified by a code. Common codes include the following:

- **L** - Identifies the address assigned to a router interface.
- **C** - Identifies a directly connected network.
- **S** - Identifies a route that is manually created to reach a specific network.
- **O/R/D, etc** - Identifies a dynamically learned network from another router
- ***** - This route is a candidate for a default route.

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Directly Connected Networks

```
RTR-Main>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
...

     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.12/32 is directly connected, GigabitEthernet0/0
```

- Networks to which one of a router's interfaces is connected are called **directly connected networks** or **directly connected routes**.
  - Automatically added when an interface is configured with an IP address and is activated.
  - Denoted by a status code of **C** in the routing table.
- The routing table also contains a **local route** for each of its **directly** connected networks,
  - indicated by the status code of **L** in the routing table
  - Used by a router to efficiently differentiate when it receives a packet for it own interface instead of a packet that needs to be forwarded.
  - Uses a prefix length of /32 for IPv4 and /128 for IPv6 addresses.
- Directly connected and Local routes appear in the routing table with the Exit Interface only

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Remote Routes

```
RTR-Main>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
...
     192.168.2.0/25 is subnetted, 2 subnets
 192.168.2.0/25 is subnetted, 2 subnets
S       192.168.2.0/25 [1/0] via 192.168.1.2, GigabitEthernet0/1.10
S       192.168.2.128/25 [1/0] via 192.168.1.2, GigabitEthernet0/1.10
     192.168.3.0/26 is variably subnetted, 2 subnets, 2 masks
D       192.168.3.0/26 [90/30720] via 192.168.1.13, 00:00:01, GigabitEthernet0/0
```

A router sources routes to remote networks using 2 methods:

- **Static routes** are manually configured paths between two networking devices
  - Denoted by a status code of **S** in the routing table.
  - May contain next hop, exit Interface or both information depending on how they were configured by the device administrator
- **Dynamic routes** are routes learned automatically from other routers using routing protocols
  - Denoted by a status code assigned to the routing protocol used to learn the route - e.g. R, O, D, I, etc.
  - Contain next hop, exit interface and timestamp information

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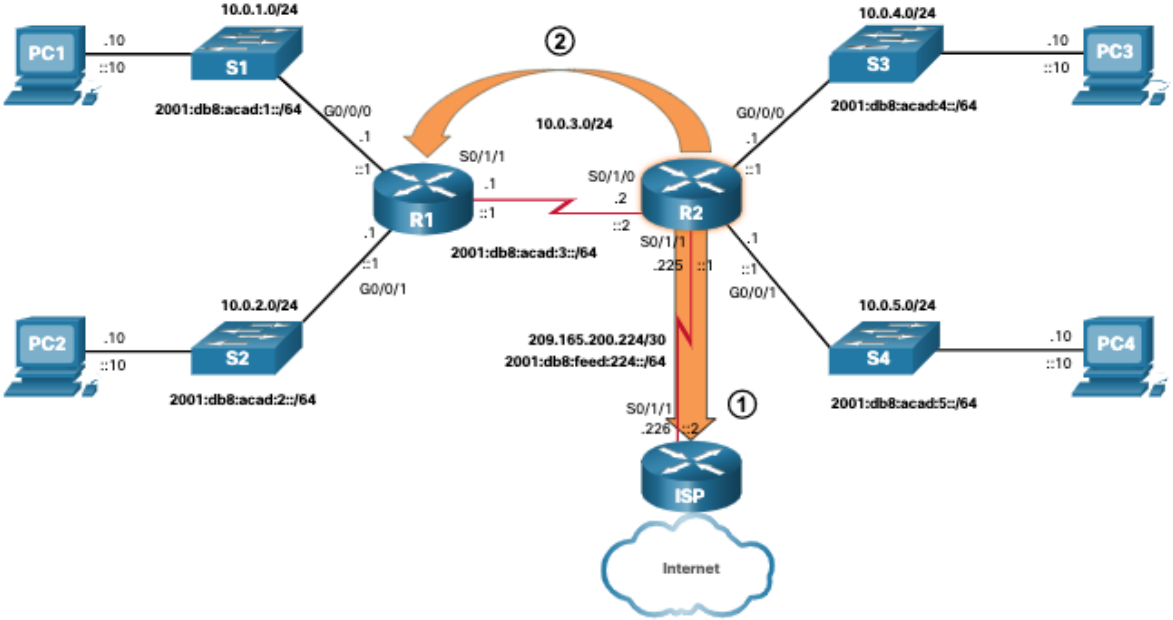
Default Route

- The **default route** specifies a next-hop router to use when the routing table does not contain a specific route that matches the destination IP address.
- Can be either a static route or learned automatically from a dynamic routing protocol.

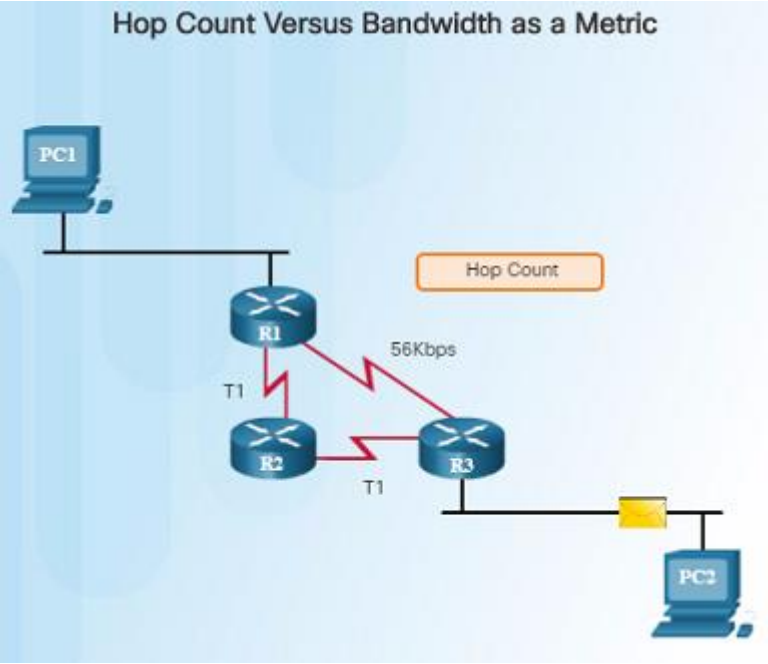


- Uses a status code with an asterisk symbol ‘*’ and has an IPv4 route entry of 0.0.0.0/0 or an IPv6 route entry of ::/0.

```
RTR-Main>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
...
D        192.168.3.64/27 [90/30720] via 192.168.1.13, 00:00:01, GigabitEthernet0/0
S*    0.0.0.0/0 [1/0] via 192.168.1.1, GigabitEthernet0/0
```

Route Metric

- If a router has multiple paths to the same destination network, it must choose the best one to place in its routing table
- Dynamic routes are assigned a **metric**, a quantitative representation of its distance to the destination network automatically calculated by the router based on the rules of the routing protocol it uses
- The path with the lowest calculated metric is considered the best path




```
RTR-Main>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
...
192.168.3.0/26 is variably subnetted, 2 subnets, 2 masks
D       192.168.3.0/26 [90/30720] via 192.168.1.13, 00:00:01, GigabitEthernet0/0
```

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Administrative Distance

- It is possible that the routing table learns about the same destination network address from more than one routing source.
- In such cases, the metric cannot be used to directly compare the routes and the **administrative distance (AD)** is used to determine route to install into the IP routing table.
  - Represents the "trustworthiness" of the route source.
  - Lower AD = more trustworthy route source.

```
RTR-Main>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
...
     192.168.2.0/25 is subnetted, 2 subnets
 192.168.2.0/25 is subnetted, 2 subnets
S       192.168.2.0/25 [1/0] via 192.168.1.2, GigabitEthernet0/1.10
S       192.168.2.128/25 [1/0] via 192.168.1.2, GigabitEthernet0/1.10
     192.168.3.0/26 is variably subnetted, 2 subnets, 2 masks
D       192.168.3.0/26 [90/30720] via 192.168.1.13, 00:00:01, GigabitEthernet0/0
```

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Administrative Distance (Cont.)

The table lists various routing protocols and their associated ADs.

Dynamic route sources

Route Source	Administrative Distance
Directly connected	0
Static route	1
EIGRP summary route	5
External BGP	20
Internal EIGRP	90
OSPF	110
IS-IS	115
RIP	120
External EIGRP	170
Internal BGP	200

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Metric vs AD

- When choosing among several route options to a destination, a router prioritizes the lowest AD followed by the lowest metric
- Example: Assume routes A to F below can all be used to reach 192.168.1.0/24

Route	Source	Administrative Distance	Metric	Next Hop
A	OSPF	110	200	10.10.1.1
B	OSPF	110	125	10.10.1.2
C	EIGRP	90	206247	10.10.1.2
D	EIGRP	90	30013	10.10.1.1
E	RIP	120	3	10.10.1.1
F	RIP	120	2	10.10.1.2

This route is placed in the routing table

Lowest AD

Lowest metric among lowest AD routes

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Routing Table Principles

There are three principles governing routing tables:

Principle 1: “Every router makes its decision alone, based on the information it has in its own routing table.”

Principle 2: “The fact that one router has certain information in its routing table does not mean that other routers have the same information.”

Principle 3: “Routing information about a path from one network to another does not provide routing information about the reverse or return path.”

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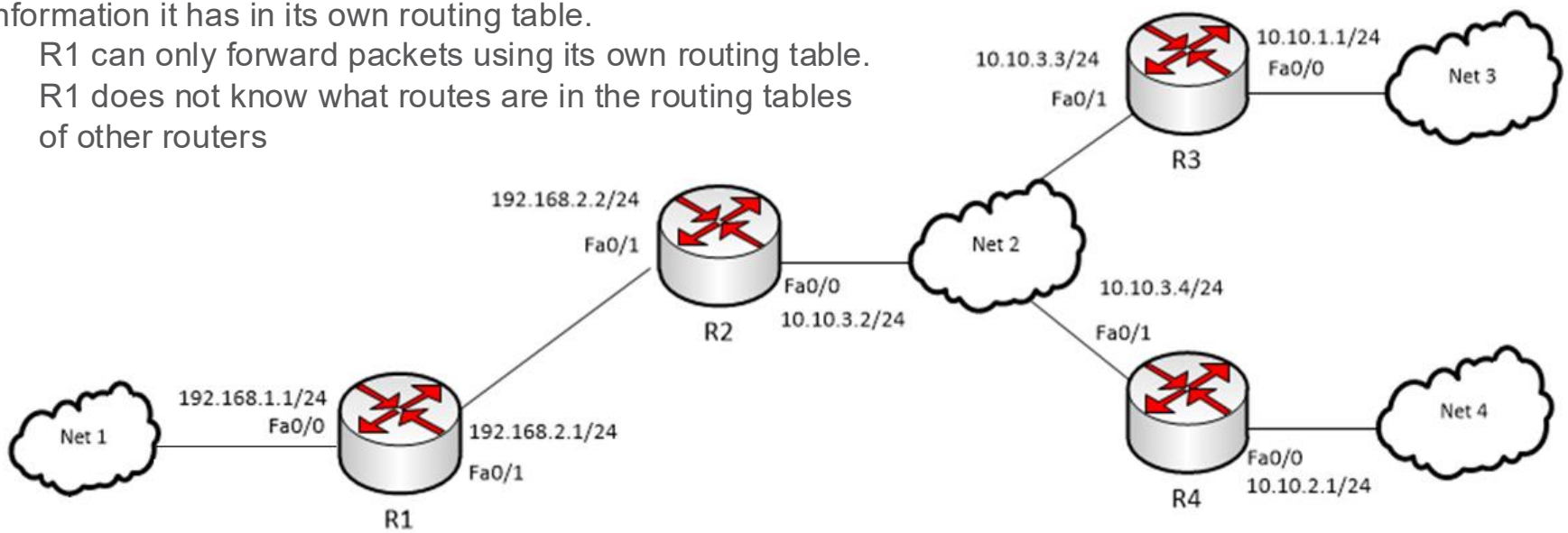
Routing Table Principle 1

Every router makes its decision alone, based on the information it has in its own routing table.

- R1 can only forward packets using its own routing table.
- R1 does not know what routes are in the routing tables of other routers



Destination	Next Hop	Exit
192.168.1.0/24	N/A	Fa0/0
192.168.2.0/24	N/A	Fa0/1
10.10.1.0/24	192.168.2.2	Fa0/1
10.10.2.0/24	192.168.2.2	Fa0/1
10.10.3.0/24	192.168.2.2	Fa0/1

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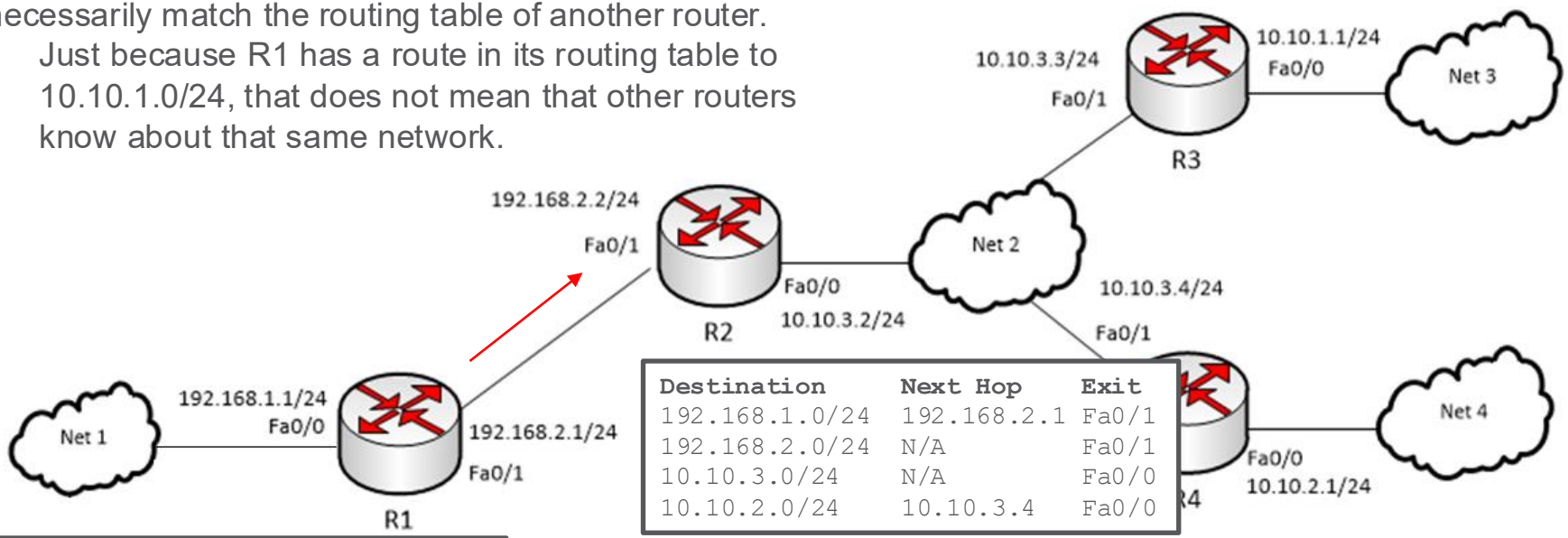
Routing Table Principle 2

The information in a routing table of one router does not necessarily match the routing table of another router.

- Just because R1 has a route in its routing table to 10.10.1.0/24, that does not mean that other routers know about that same network.



Destination	Next Hop	Exit
192.168.1.0/24	N/A	Fa0/0
192.168.2.0/24	N/A	Fa0/1
10.10.1.0/24	192.168.2.2	Fa0/1
10.10.2.0/24	192.168.2.2	Fa0/1
10.10.3.0/24	192.168.2.2	Fa0/1

Destination	Next Hop	Exit
192.168.1.0/24	192.168.2.1	Fa0/1
192.168.2.0/24	N/A	Fa0/1
10.10.3.0/24	N/A	Fa0/0
10.10.2.0/24	10.10.3.4	Fa0/0

No route to 10.10.1.0, still can't reach the destination network

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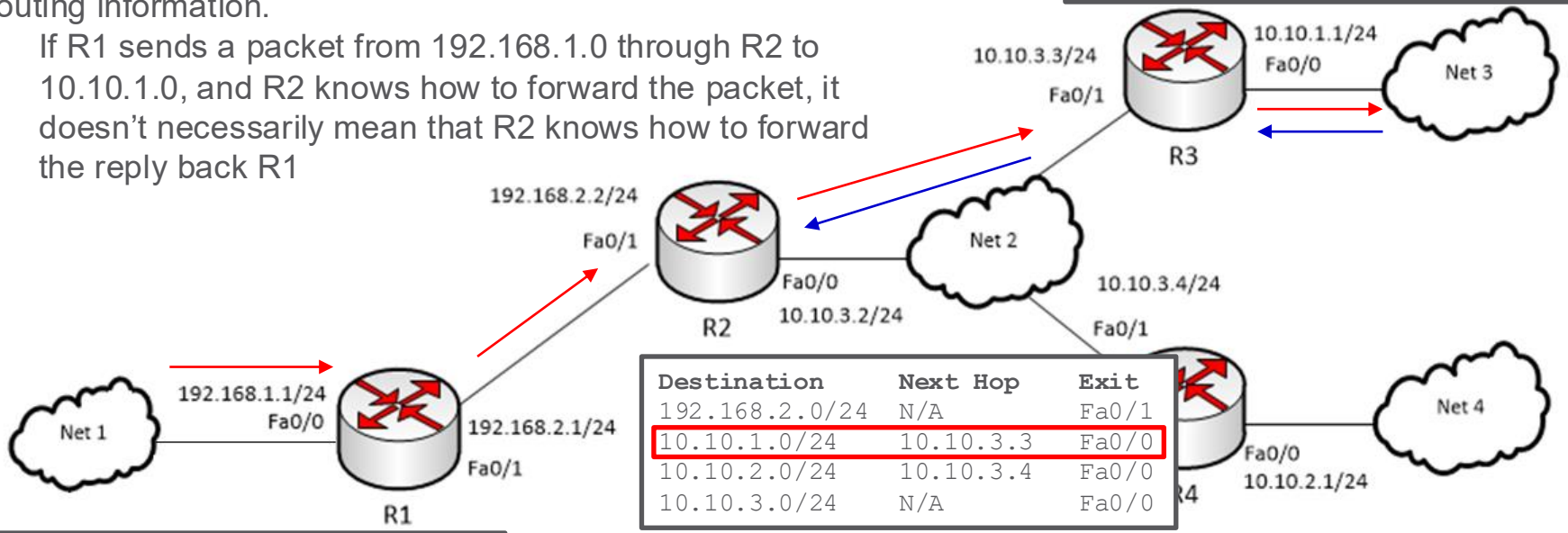
Routing Table Principle 3

Routing information about a path does not provide return routing information.

- If R1 sends a packet from 192.168.1.0 through R2 to 10.10.1.0, and R2 knows how to forward the packet, it doesn't necessarily mean that R2 knows how to forward the reply back R1

Destination	Next Hop	Exit
192.168.1.0/24	10.10.3.2	Fa0/1
192.168.2.0/24	10.10.3.2	Fa0/1
10.10.1.0/24	N/A	Fa0/0
10.10.2.0/24	10.10.3.4	Fa0/1
10.10.3.0/24	N/A	Fa0/1



Destination	Next Hop	Exit
192.168.2.0/24	N/A	Fa0/1
10.10.1.0/24	10.10.3.3	Fa0/0
10.10.2.0/24	10.10.3.4	Fa0/0
10.10.3.0/24	N/A	Fa0/0

Destination	Next Hop	Exit
192.168.1.0/24	N/A	Fa0/0
192.168.2.0/24	N/A	Fa0/1
10.10.1.0/24	192.168.2.2	Fa0/1
10.10.2.0/24	192.168.2.2	Fa0/1
10.10.3.0/24	192.168.2.2	Fa0/1

No route back to 192.168.1.0.
Reply fails

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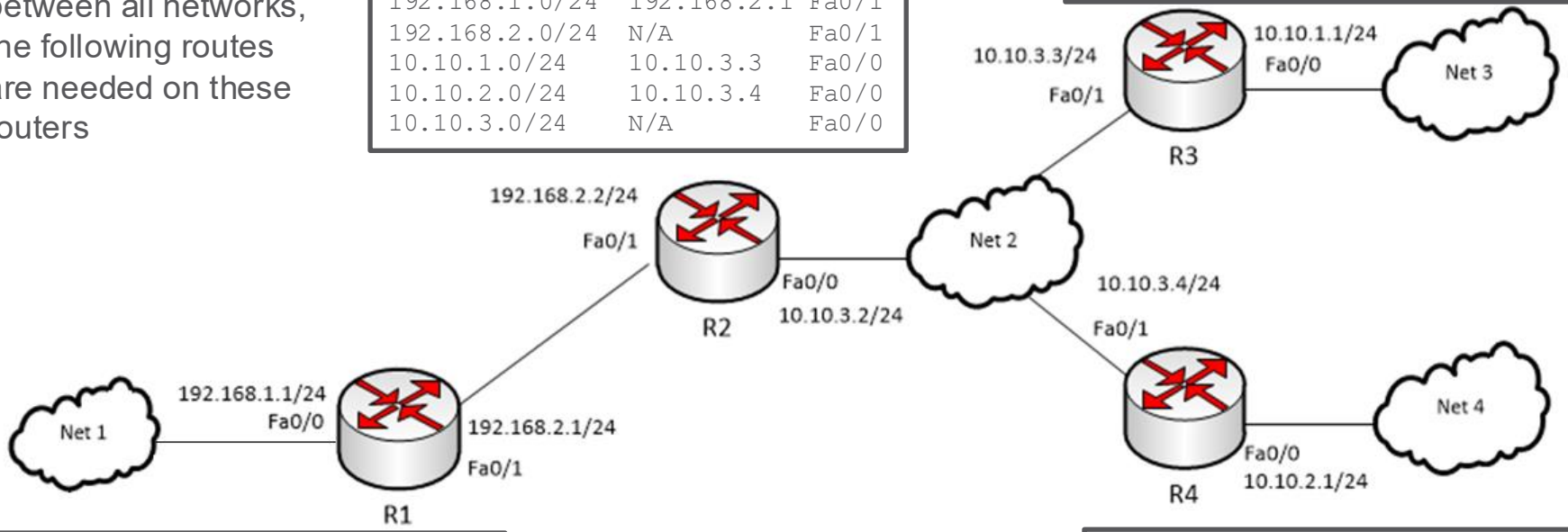
Routing Table Principles

For full connectivity between all networks, the following routes are needed on these routers

Destination	Next Hop	Exit
192.168.1.0/24	192.168.2.1	Fa0/1
192.168.2.0/24	N/A	Fa0/1
10.10.1.0/24	10.10.3.3	Fa0/0
10.10.2.0/24	10.10.3.4	Fa0/0
10.10.3.0/24	N/A	Fa0/0

Destination	Next Hop	Exit
192.168.1.0/24	10.10.3.2	Fa0/1
192.168.2.0/24	10.10.3.2	Fa0/1
10.10.1.0/24	N/A	Fa0/0
10.10.2.0/24	10.10.3.4	Fa0/1
10.10.3.0/24	N/A	Fa0/1



Destination	Next Hop	Exit
192.168.1.0/24	N/A	Fa0/0
192.168.2.0/24	N/A	Fa0/1
10.10.1.0/24	192.168.2.2	Fa0/1
10.10.2.0/24	192.168.2.2	Fa0/1
10.10.3.0/24	192.168.2.2	Fa0/1

Destination	Next Hop	Exit
192.168.1.0/24	10.10.3.2	Fa0/1
192.168.2.0/24	10.10.3.2	Fa0/1
10.10.1.0/24	10.10.3.3	Fa0/1
10.10.2.0/24	N/A	Fa0/0
10.10.3.0/24	N/A	Fa0/1

Structure of an IPv4 Routing Table

- IPv4 was standardized using the now obsolete classful IP addressing architecture.
- The IPv4 routing table of Cisco routers is organized using this classful structure and retains this even if classful IP addressing is no longer practiced.
- A destination network is a subnet of a classful network is called a **child route** and appears indented in the routing table

```
RTR-Main>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
...

Gateway of last resort is 192.168.1.1 to network 0.0.0.0

     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.12/32 is directly connected, GigabitEthernet0/0
     192.168.2.0/25 is subnetted, 2 subnets
S       192.168.2.0/25 [1/0] via 192.168.1.2, GigabitEthernet0/1.10
S       192.168.2.128/25 [1/0] via 192.168.1.2, GigabitEthernet0/1.10
```

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Structure of an IPv4 Routing Table

- IPv4 was standardized using the now obsolete classful IP addressing architecture.
- The IPv4 routing table of Cisco routers is organized using this classful structure and retains this even if classful IP addressing is no longer practiced.
- A classful network appearing above a child route is called a **parent route.** It indicates which classful network the child route was subnetted from.

```
RTR-Main>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
...

Gateway of last resort is 192.168.1.1 to network 0.0.0.0

     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.12/32 is directly connected, GigabitEthernet0/0
     192.168.2.0/25 is subnetted, 2 subnets
S       192.168.2.0/25 [1/0] via 192.168.1.2, GigabitEthernet0/1.10
S       192.168.2.128/25 [1/0] via 192.168.1.2, GigabitEthernet0/1.10
```

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Structure of an IPv6 Routing Table

- The concept of network classes was never part of IPv6, so the structure of an IPv6 routing table is very straight forward.
- Every IPv6 route entry is formatted and aligned the same way even if subnetting is implemented in an IPv6 network

```
R1# show ipv6 route
Codes: L - local, C - connected, S - static, R -
...
OE2 ::/0 [110/1], tag 2
   via FE80::2:C, Serial0/0/1
C 2001:DB8:ACAD:1::/64 [0/0]
   via GigabitEthernet0/0/0, directly connected
L 2001:DB8:ACAD:1::1/128 [0/0]
   via GigabitEthernet0/0/0, receive
C 2001:DB8:ACAD:2::/64 [0/0]
  via GigabitEthernet0/0/1, directly connected
L 2001:DB8:ACAD:2::1/128 [0/0]
   via GigabitEthernet0/0/1, receive
C 2001:DB8:ACAD:3::/64 [0/0]
   via Serial0/1/1, directly connected
L 2001:DB8:ACAD:3::1/128 [0/0]
   via Serial0/1/1, receive
O 2001:DB8:ACAD:4::/64 [110/50]
   via FE80::2:C, Serial0/1/1
O 2001:DB8:ACAD:5::/64 [110/50]
   via FE80::2:C, Serial0/1/1
L FF00::/8 [0/0]
   via Null0, receive
R1#
```

7.2 Static and Dynamic Routing

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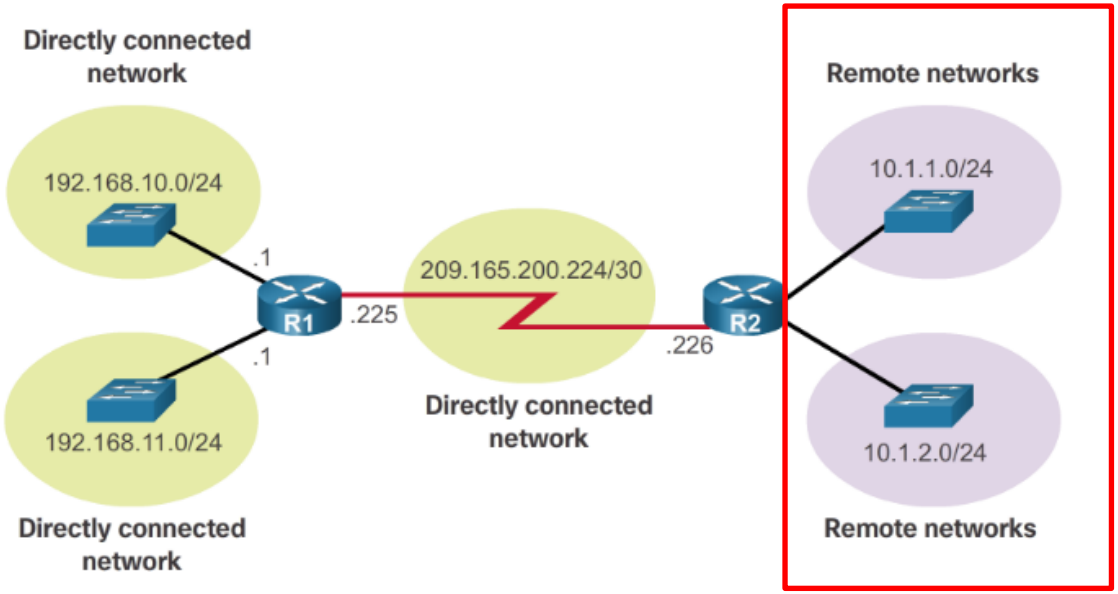
Reaching Remote Networks

- **Recall**: To reach remote networks, routers must use remote routes. These are either:
  - Statically configured routes or
  - Dynamic routes learned using a routing protocol

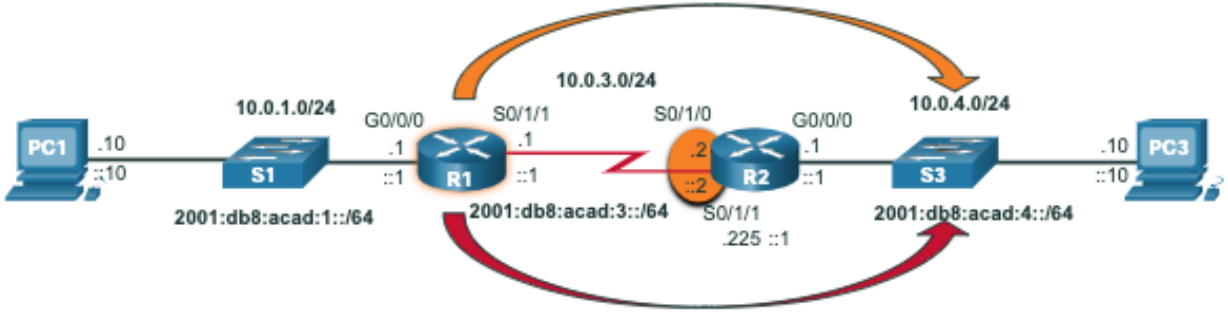


Static Routing

- Static routes are manually configured by a network administrator to define an explicit path between two networking devices.
- Are not automatically updated and must be manually reconfigured if the network topology changes.

```
R1 (config)# ip route 10.0.4.0 255.255.255.0 10.0.3.2
```

Remote IPv4 network address — IPv4 address of next hop router

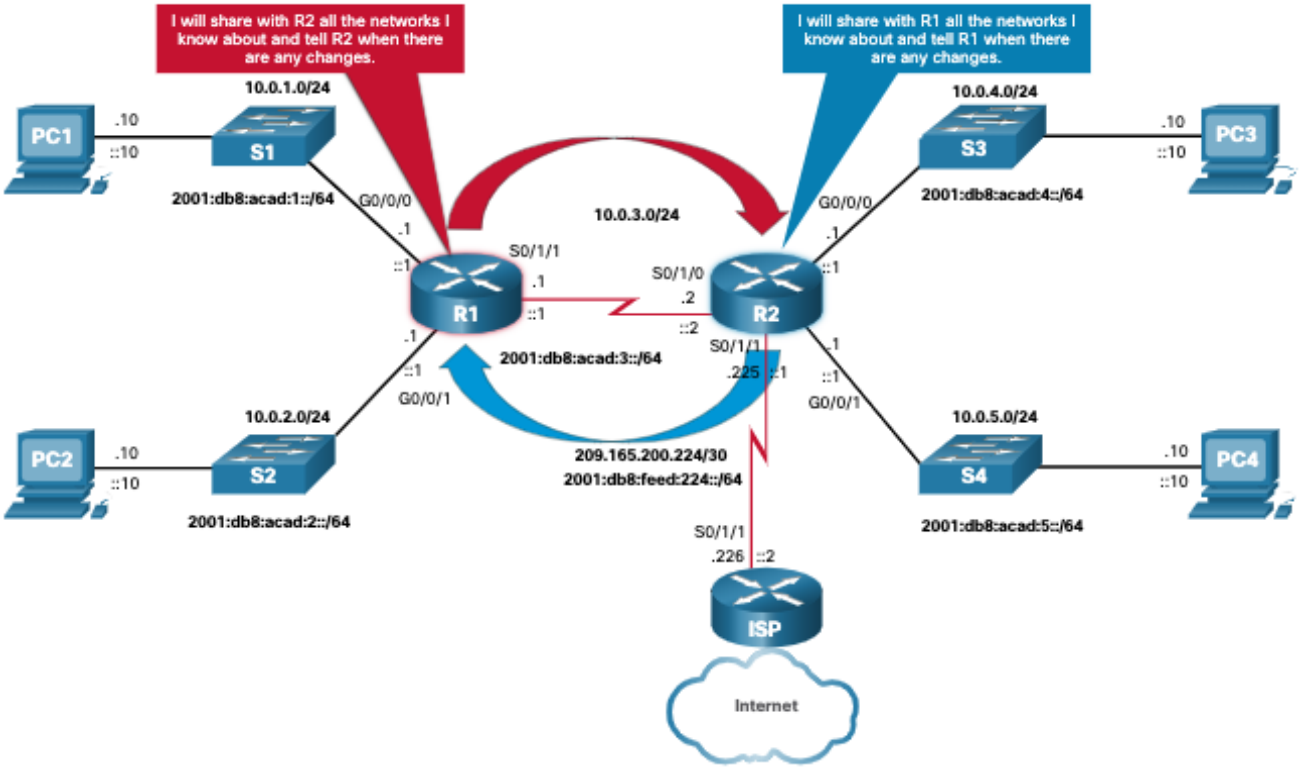



```
R1 (config)# ipv6 route 2001.db8.acad.4::/64 2001.db8.acad.3::2
```

Remote IPv6 network address — IPv6 address of next hop router

Dynamic Routing

Dynamic routing uses routing protocols so that routers automatically share information among themselves about the reachability and status of remote networks.



Dynamic Routing Protocol

- A **routing protocol** is used exchange routing information and populate the routing table with the choice of best paths.
- The purpose of dynamic routing protocols includes the following:
  - Discovery of remote networks
  - Maintaining up-to-date routing information
  - Choosing the best path to destination networks
  - Ability to find a new best path if the current path is no longer available

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7.3 Routing Operation

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Matching Packets to Routes

- Recall: A route destination is represented by a prefix (network address), and prefix length (subnet mask)
- To determine if a route is suitable for forwarding, a packet destination must match the route prefix by a number of far-left bits that is equal to or more than the prefix length
- Example:

Must match at least 20 far-left bits to use this route

Route Prefix =	172.16.16.0/20	10101100.00010000.00010000.00000000	
Packet A Dest =	172.16.25.93	10101100.00010000.00011001.01011100	✓ Match
Packet B Dest =	172.16.18.3	10101100.00010000.00010010.00000011	✓ Match
Packet C Dest =	172.16.42.250	10101100.00010000.00101010.11111010	X No Match

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Longest Match

When the routing table contains several routes that fulfill the prefix match requirement for a packet, the preferred route is the one with the longest matched prefix to the packet destination

Example:

Destination IPv4 Address		Address in Binary
172.16.25.93		10101100.00010000.00011001.01011100
Route Entry	Prefix/Prefix Length	Address in Binary
1	172.16.0.0/**16**	10101100.00010000.00000000.00000000
2	172.16.25.0/**24**	10101100.00010000.00011001.00000000
3	172.16.25.64/**26**	10101100.00010000.00011001.01000000 ✓ Longest Match

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Longest Match

When the routing table contains several routes that fulfill the prefix match requirement for a packet, the preferred route is the one with the longest matched prefix to the packet destination

Example:

Destination	**2001:db8:c000::99/48**	
Route Entry	Prefix/Prefix Length	Does it match?
1	**2001:db8:c0**00::**/40**	Match of 40 bits
2	**2001:db8:c000**::**/48**	Match of 48 bits (longest match)
3	**2001:db8:c000**:5555::**/64**	Does not match 64 bits

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Practice

Given the following routing table:

Route	Prefix/Prefix Length	Binary Equivalent
A	192.168.0.0/22	11000000.10101000.00000000.00000000
B	192.168.1.0/25	11000000.10101000.00000001.00000000
C	192.168.1.128/26	11000000.10101000.00000001.10000000
D	0.0.0.0/0	00000000.00000000.00000000.00000000

Which route will be used to forward the following packets?

Destination	Binary Equivalent	Which Route?
192.168.3.26	11000000.10101000.00000011.10000000	A
192.168.1.130	11000000.10101000.00000001.10000010	C
192.168.1.250	11000000.10101000.00000001.11111010	A
192.168.5.63	11000000.10101000.00000101.00000000	D

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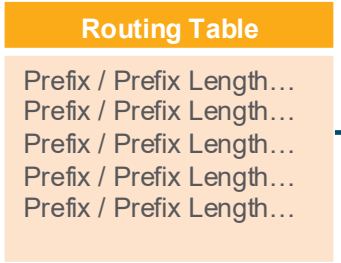
Packet Forwarding Decision Process

1. A data frame with an encapsulated IP packet arrives through the ingress interface.

2. The router examines the destination IP address consults its routing table

3. The router finds the longest matching prefix in the routing table



4a. If no match is found, packet is discarded

4b. If a match is found, router encapsulates packet with appropriate header for egress interface

5a. Packet is for a directly connected network

5b. Packet is for a remote network

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Packet Forwarding Decision Process (Cont.)

After a router has performed a route lookup, it could do the following:

- Forward the packet to a device on a directly connected network
  - If the route entry indicates that the egress interface is a directly connected network, the packet can be forwarded directly to the destination device. Typically this is an Ethernet LAN.
  - To encapsulate the packet in the Ethernet frame, the router determines the destination MAC address associated with the destination IP address of the packet through ARP (IPv4) or Neighbor Discovery (IPv6)
- Forward the packet to a next-hop router
  - If the route entry indicates that the destination IP address is on a remote network, the packet must be forwarded to the next-hop router indicated in the route entry.
  - If the forwarding router and the next-hop router are on an Ethernet network, a similar process (ARP and ICMPv6 Neighbor Discovery) will occur for determining the destination MAC address of the packet as described previously. A different process is used for other network types
- Drop the Packet if there is no match among available routes

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Load Balancing

```
RTR-Main>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
...

Gateway of last resort is 192.168.1.1 to network 0.0.0.0

     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
R    192.168.2.0/24 [120/1] via 192.168.1.3, 00:00:18, GigabitEthernet0/0
                    [120/1] via 192.168.1.2, 00:00:08, GigabitEthernet0/0
     192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, GigabitEthernet0/1
L       192.168.3.1/32 is directly connected, GigabitEthernet0/1
```

- When the router has multiple equal cost paths to a single destination network, it balances the packet forwarding load among the equal cost paths to increase throughput
- Equal cost load balancing is implemented automatically by dynamic routing protocols. It is enabled with static routes when there are multiple static routes to the same destination network using different next-hop routers.

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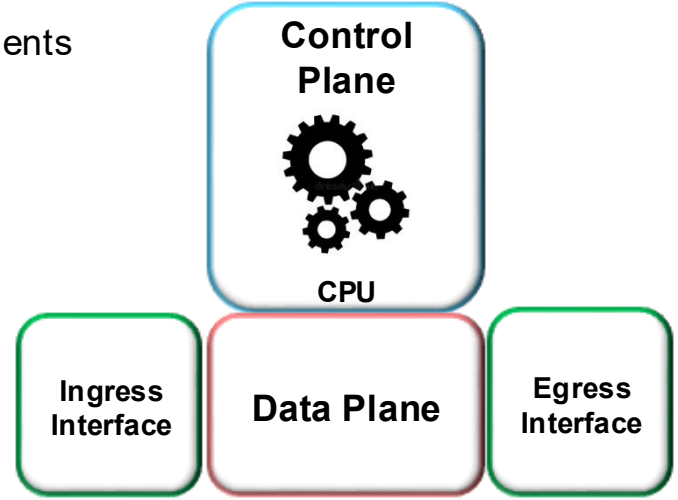
Router Architecture

- The typical router architecture separates its functions into 2 components
  - The **Control Plane** performs the decision-making processes to determine how packets should be forwarded
    - Routing protocol processes and calculations
    - Building the routing table
  - The **Data Plane** performs the actual forwarding of packets
    - Extracting the destination address
    - Routing table lookups
    - Encapsulate packets for the outgoing interface



- Cisco routers support the following packet forwarding mechanisms which differ by how both planes work together to forward packets to their destinations
  - Process switching
  - Fast switching
  - Cisco Express Forwarding (CEF)

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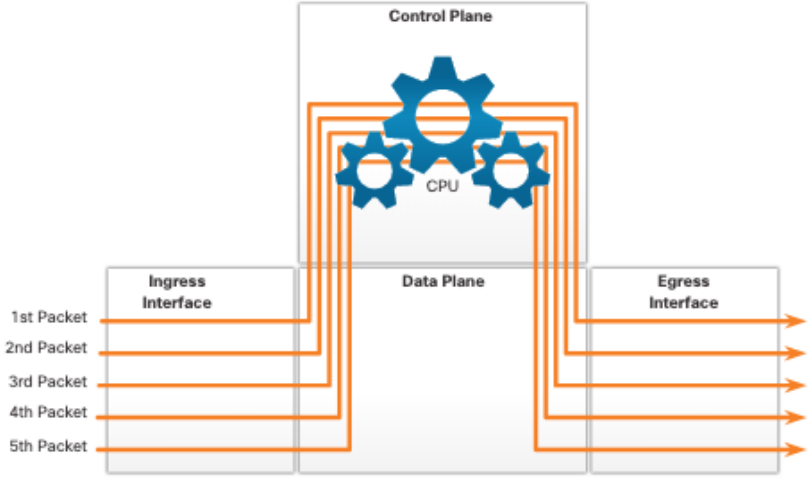
Packet Forwarding Mechanisms

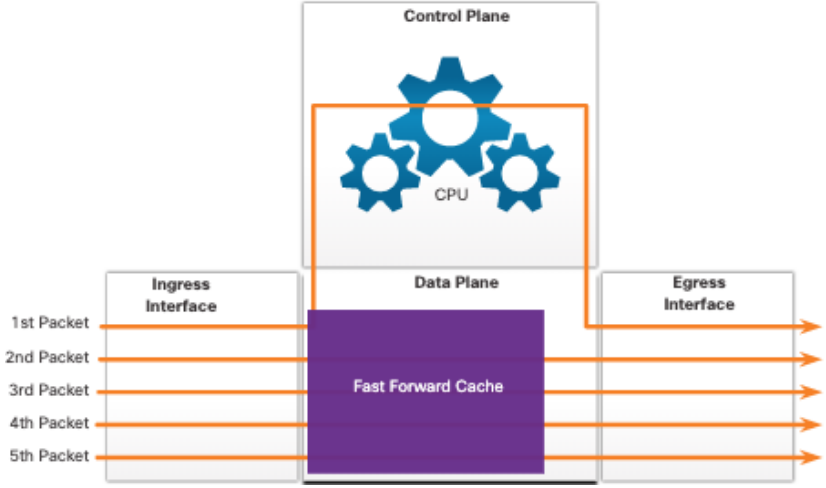
Process Switching:

- An older packet forwarding mechanism
- For **each** packet that arrives on an interface, it is forwarded to the control plane where the CPU matches the destination address with an entry in its routing table
- Least efficient



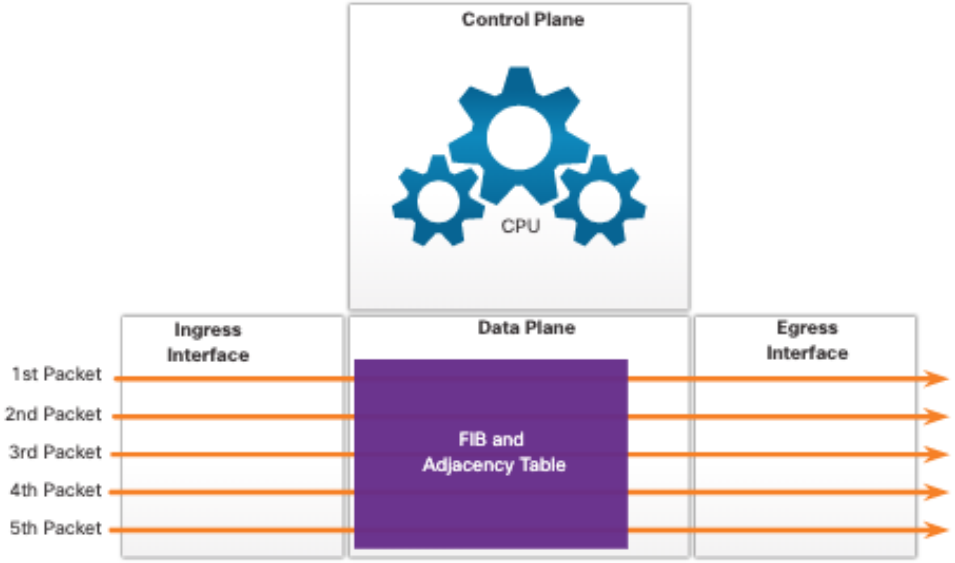
Fast Switching:

- Successor to process switching
- CPU uses process switching to forward the first packet to a newly encountered destination and caches the route
- Subsequent packets to the same destination can use the cached information without needing to repeat the route lookup process



Packet Forwarding Mechanisms (Cont.)

- **Cisco Express Forwarding (CEF):**
  - Most recent and default packet-forwarding mechanism for Cisco routers.
  - Builds a precalculated Forwarding Information Base (FIB), and an adjacency table which contain all the information that a router would have to consider when forwarding a packet to reduce routing table lookups .
  - FIB and adjacency table are updated when a network topology change occurs
  - Highest throughput approach available on Cisco routers



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Questions?

What did you learn in this module?

- A router may be used to forward packets to directly connected and remote networks
- Each router maintains a routing table – a data structure stored in RAM to track the list of known networks and how to reach them
- Each routing table entry consists of a
  - Source – How route was acquired
  - Prefix and prefix length – destination network
  - Administrative distance – route source reliability, lower is better
  - Metric – route distance, lower is better
  - Next Hop – neighbor router that can forward packets to destination network
  - Exit interface – egress interface to use to forward the packet to its destination

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What did you learn in this module?

- Directly connected routes are automatically learned when an interface is active and is configured with an IP address
- Remote routes are learned through
  - Static routing - Manual route configuration and does not automatically adapt to network topology changes. It is suitable for small networks
  - Dynamic routing - Routers communicate with each other to learn networks and routes using a routing protocol. It automatically adapts to topology changes and is suitable for large networks
- Each router follows the 3 routing principles:
  - Independent decision making based on own routing table
  - Presence of a route does not guarantee the presence of the same route on other routers
  - A route from one network to another does not provide the return path
- Routing protocols are composed of data structures, routing messages and algorithms.

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What did you learn in this module?

- Routers determine packet forwarding packets by selecting the route with the longest matched prefix / prefix length to the packet destination address
- When forwarding packets, routers may
  - Send packets to the destination host on a directly connected network
  - Send packets to the next hop for a remote network
  - Drop packets without a matching route in its routing table
- Cisco routers offer 3 options for packet forwarding mechanisms
  - Process switching – Lookup routing table for each packet
  - Fast switching – Lookup routing table for first packet to destination, remember path for succeeding packets to same destination using a cache
  - Cisco express forwarding – precompute a forwarding information base according to the routing table so no need to lookup routing table for packets.

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